AI Gateways for EuroHPC AI Factories and Antennas: Functions, Tools and the SAIFA Example

Aleksandra Stojanović
School of Electrical Engineering
University of Belgrade
Belgrade, Serbia
sa230234d@student.etf.bg.ac.rs

Dimitrije Pešić
School of Electrical Engineering
University of Belgrade
Belgrade, Serbia
pd230014d@student.etf.bg.ac.rs

Zaharije Radivojević
School of Electrical Engineering
University of Belgrade
Belgrade, Serbia
zaki@etf.rs

***Abstract*—The EuroHPC AI Factories initiative has built a network of AI supercomputing centres in Europe, and AI Factory Antennas bring their services to countries without large AI infrastructure. As a result, to run one experiment, users may need several accounts, applications and portals, from EuroHPC access calls and the EuroHPC Federation Platform to Factory offices, national Antennas and the AI-on-Demand Platform. This paper gives an overview of these access points and proposes a general model of an *AI gateway*, the part of an AI Factory or Antenna that users interact with and through which they get computing power, work with data, models and workflows, and receive training and support. The main contribution is a clear division of responsibilities: what the gateway does itself, what it leaves to existing systems, and which system makes the final decision on access, resource allocation and accounting. We define ten groups of functions, compare them with three existing solutions, Waldur, Open OnDemand and the EuroHPC Federation Platform, and list possible open-source tools. As an example, we use the gateway of the Serbian AI Factory Antenna (SAIFA), linked with Pharos and IT4LIA.**

***Index Terms*—AI Factory, AI Factory Antenna, AI gateway, EuroHPC, federation, high-performance computing, MLOps, science gateway.**

I. Introduction

Modern artificial intelligence (AI) research needs large computing resources. To give such resources to more users, the EuroHPC Joint Undertaking (JU) started the AI Factories initiative. By October 2025, 19 AI Factories had been selected. Each of them has an AI supercomputer and offers services such as data management, help with model development, training courses and expert support [1], [2]. In October 2025, 13 AI Factory Antennas were also selected to bring these services to more countries. Each Antenna is linked with one or more AI Factories [3], [4].

However, these services are reached through many separate access points, which makes it harder for users to get their work done. Startups, small and medium-sized enterprises (SMEs), researchers and public institutions have to use several portals, accounts, application procedures and service lists before they can run one experiment. Science gateways solved a similar problem for grid and high-performance computing (HPC) resources by giving users simple web access [5], [6]. AI work needs more than this: users go through the whole process of building a model, share data and models safely and use large language models (LLMs). Many of them also have to learn how to use HPC and AI tools first, and they often work with both national and European sites.

The main contribution of this paper is a general model of an AI gateway. The model states which functions the gateway performs itself and which it leaves to existing systems, such as schedulers, storage systems or partner Factories. It also states which system makes the final decision on access, resource allocation and accounting. The same model can be used by an AI Factory and by an Antenna. An Antenna has one extra task: to connect its users with one or more linked Factories. Section II reviews the current access points, Sections III and IV present the model and its functions, Section V compares it with existing solutions, and Section VI presents the gateway of the Serbian AI Factory Antenna (SAIFA) [7], [8] as an example.

II. How Users Access AI Factories Today

A. AI Factories

AI Factories are run by national host organisations around EuroHPC supercomputers. The JU lists the services they offer [9]: expert help and consulting, computing time on AI-optimised supercomputers and cloud resources, data storage and management, co-working spaces, training, and AI tools, software and workflows. Computing time is given through EuroHPC access calls in four modes [10]: *Playground* and *Fast Lane* access for startups and SMEs, *Large Scale* access for work that needs more than 50,000 GPU hours, and *AI for Science and Collaborative EU Projects* for scientific and public sector users. Each Factory also has its own contact point and its own rules for users. For example, Pharos in Greece has a Front Desk as the central contact point for users [11].

B. AI Factory Antennas

Antennas extend the services of existing AI Factories [4]. An Antenna usually offers a national entry point, smaller local computing resources, AI test environments, help for specific sectors and training. For large jobs, which need more than these local resources, an Antenna uses the resources of its

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linked Factories. One Factory can work with several Antennas and vice versa: Pharos works with four Antennas (Pharos-CY, CALYPSO, VEZILKA and SAIFA), and the Swiss Antenna HEARTS with four Factories [3].

C. Shared European platforms

The first version of the EuroHPC Federation Platform (EFP) was released in April 2026. It gives users one entry point and one login for several EuroHPC systems. There, they can also request resources, transfer data and find software in a shared catalogue. AI Factories are not yet part of it and will be connected in later versions [12]. The AI-on-Demand Platform is a European catalogue where users can find AI resources [13]. Table I summarises all access points.

TABLE I
ACCESS POINTS IN THE AI FACTORY NETWORK

Access point	What it offers to the user
EuroHPC access calls	Applying for computing time (four modes for industry and science)
EuroHPC Federation Platform	One login, common resource allocation and data transfer, shared software catalogue for EuroHPC systems
AI Factory contact office	Consulting, expert help, training, data services and computing at a national site
AI Factory Antenna	National entry point, local computing, help for new users, sending users and results to linked Factories
AI-on-Demand Platform	European catalogue of AI resources

These access points cover different needs, but none of them offers all these services in one place together with access to the European services. This is the task of the AI gateway.

III. General Model of an AI Gateway

An AI gateway works as *middleware*. It receives requests from users or applications, checks all of them against common rules for identity, access and metadata, and routes each request to the appropriate service. Fig. 1 shows its main parts, from login and the catalogue to quotas, audit and connections to partners. The actual work is still done by existing systems, such as schedulers, storage systems, model servers, learning platforms and partner platforms.

The ten function groups in Table II come from the SAIFA design. Three rules apply to all of them: every action is checked against the user's permissions, every dataset and model has a version and a record of its origin (lineage), and every job is checked against the quota before it is queued.

The split of responsibilities follows the SAIFA design [8]. The institutional or EuroHPC identity provider authenticates the user, and the gateway links this identity to its own account. Within a project, the project owner invites members and assigns roles; within an organisation, a representative manages membership and allocated resources. A partner AI Factory authorises the use of its resources, while the gateway only passes the user's identity on to the partner and tracks usage for each Factory. For local resources, quota and accounting logic

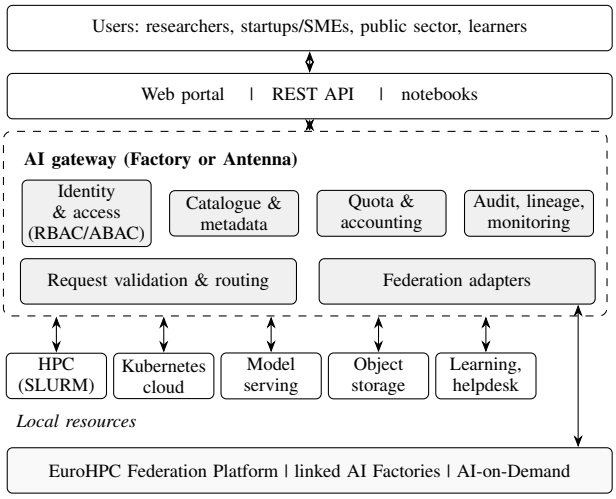


Fig. 1. General model of an AI gateway between users, local resources and European AI Factory services.

TABLE II
MAIN FUNCTION GROUPS AND THEIR COVERAGE BY EXISTING SOLUTIONS

Group	Main features	W	O	E
Identity & access	Institutional (eduGAIN) and EuroHPC sign-in, account linking, RBAC/ABAC, scoped API tokens, revocation	◐	◐	●
Organisations & projects	Workspaces, membership, roles, resource visibility (private, project, platform, public)	●	○	◐
Catalogue	Search of models, datasets, workflows, services and courses; source attribution of federated entries	◐	○	◐
Data & model management	Resumable upload, versioning, model and data cards, review and publication workflow	○	◐	◐
Compute jobs	Job templates, quota reservation, routing to HPC or Kubernetes, status, logs, result retrieval	◐	●	●
Inference	Authenticated, rate-limited access to served models and LLMs	○	○	○
Experiments & MLOps	Experiment tracking, run comparison, benchmark evaluation, lineage across job, dataset and model	○	○	○
Training	Courses, learning paths, preconfigured notebook labs with controlled resources	○	◐	○
Operations	Monitoring, usage accounting, audit trail, notifications, contextual helpdesk tickets	◐	○	◐
Federation	Catalogue exchange, metadata mapping, identity brokering, cross-Factory execution and accounting	◐	○	◐

W: Waldur [14], O: Open OnDemand [6], [15], E: EFP [16]. ● provided, ◐ partly (configuration or extension), ○ not provided (new development).

belongs to the gateway, which keeps the record of available quota, reserves quota before a job is queued and records inference consumption. The gateway stores metadata, review decisions, job records, audit logs and lineage, but job status always comes from the scheduler, locally or at the partner site. Some details, such as the quota hierarchy, review states and the split of tasks with system operators, are not final yet [8].

IV. Description of the Main Functions

A. Login, access and collaboration

Users log in with their university or institute account (for example through eduGAIN) or with a EuroHPC account.

TABLE III
POSSIBLE TOOLS FOR EACH PART AND WORK NEEDED FOR THE GATEWAY, BASED ON [8]

Part	Main option(s)	Work needed for the gateway
User access		
Portal & UX	Next.js/React	Unified UI, backend-for-frontend, tool embedding
Identity & policies	Keycloak; Open Policy Agent	Role mapping, ABAC attributes, account linking, policy model
Notebooks	JupyterHub + KubeSpawner	Launch from portal, quotas, dataset mount, audit
Data and models		
Catalogue & search	CKAN or Dataverse; OpenSearch + pgvector	Unified catalogue, access filtering in index, federation
Model registry & experiments	MLflow	Model cards, approval workflow, access rights
Storage	PostgreSQL, S3 storage on MinIO/Ceph, Valkey	Metadata links, lifecycle, retention
Lineage	OpenLineage + Marquez	Links between jobs, datasets, models and results
Compute and inference		
LLM inference & serving	vLLM; KServe	API proxy, quotas, routing, audit
Fine-tuning	Hugging Face TRL/PEFT	Approved templates, HPC submission, result registration
HPC & batch	SLURM, Open OnDemand; Kubernetes Jobs	HPC bridge, unified job model, SLURM/Kubernetes routing
Workflows & containers	Argo Workflows; Harbor + Trivy, Apptainer on HPC	Workflow templates, image approval and HPC conversion
Operations and federation		
Observability & audit	Prometheus, Grafana, Loki, OpenTelemetry; append-only log + write-once archive	Correlation with user/job, shared audit events
Learning & support	Moodle; Zammad or GLPI	Portal links, lab quotas, contextual tickets
Federation & accounting	DCAT/DCAT-AP adapters; Waldur as cross-layer option	Partner adapters, per-Factory accounting

Access is decided by the user's role (role-based access control, RBAC) and by other attributes, such as organisation, project and type of data (attribute-based access control, ABAC) [17]. Seeing a resource in the catalogue and using it are two separate permissions, and search shows resources of other projects only if they are public or their owners allow it.

B. Datasets and models

To train a model, a user only needs access to the dataset and can repeat runs as often as needed. Each result is linked automatically to its dataset, settings and job. Formal review, in which a data administrator or model reviewer checks the metadata, origin, licence and test results, is needed only when a dataset or model is published [8]. A dataset or model version is published only after the reviewer approves it, and the decision is recorded. Only published resources can be shared with partners. Large files are uploaded in parts, and each part is checked for errors individually. Once stored, a version cannot be changed. Published resources have FAIR metadata [18], which eases sharing with partner catalogues.

C. Jobs, inference and experiments

Before a job is queued, the gateway checks the input data and the user's access, and reserves quota. After that, it sends the job to HPC through SLURM [19] or to Kubernetes [20]. Sensitive data, such as medical records, does not have to be moved: the job runs on the cluster that stores the data and reads it there, without copying it elsewhere. The gateway collects status, logs and results and shows them in the same way for all systems. Containers are converted to Apptainer/Singularity images [21] for HPC. Requests to a model go through a proxy. It authenticates the user and checks the limits before forwarding the request, and records the usage after the answer. When runs are compared, a person always decides which model to keep.

D. Working with partners: Factory and Antenna

A Factory mostly *publishes* its services and resources to European platforms. An Antenna also has to *connect* its users with one or more linked Factories. It passes user identities on to partners and translates metadata into their formats. It also sends large test jobs to a partner site, brings the results back to the user's project and keeps track of usage for each Factory. Each partner is connected through its own catalogue adapter, for example using DCAT, a standard format for data catalogues [22]. The gateway shows where each partner resource comes from and never edits it. At first, the links to partner Factories only read data from them or are simulated for testing. If a partner is unavailable, the gateway still works for local users. In the future, when AI Factories join the EFP [12], gateways should reuse its login and allocation instead of building separate ones.

V. EXISTING SOLUTIONS AND POSSIBLE TOOLS

Three existing solutions already cover part of the model. Waldur is a platform for projects, allocation and accounting on cloud and HPC systems [14], and Open OnDemand is a web portal for HPC centres [6]. The EFP [12] is itself built on MyAccessID, Waldur and Open OnDemand [23]. Table II shows what each of them offers to users. These solutions can already be used for login, projects, allocation, accounting and interactive HPC work, which mainly need to be set up rather than developed. What they lack are the AI-specific functions (dataset and model review, inference with quotas, experiment lineage) and a way to send an Antenna's jobs to linked Factories while tracking usage per Factory. The model adds these functions and common rules for access, quotas and metadata, which is where most new development is needed.

Existing tools are used first, and new ones are built only when needed. Table III gives the main options for each part and the work that each gateway still has to do itself;

alternatives are listed in [8]. Most options are well known open-source projects, for example JupyterHub [24], [25], MLflow [26], vLLM [27] and Open OnDemand [6].

Tools are chosen in two steps. First, each tool must meet the basic requirements for its area, for example immediate revocation of access. Second, the tools that pass are compared by ease of use, integration and operating effort, speed, cost, maturity and legal issues (GDPR, data location, licences), preferring open-source tools that can run on local servers [8].

VI. Example: SAIFA

SAIFA, the Serbian Antenna linked with Pharos and IT4LIA, runs from May 2026 to April 2029 [7]. Its local resources are the PARADOX HPC cluster at the Institute of Physics Belgrade and the AI computing infrastructure of the Office for IT and eGovernment. The function groups in Table II come from 23 user stories, 18 from its four domains (language and culture, energy and manufacturing, health, environment) and 5 cross-domain. Examples include flood prediction with AutoML, synthetic genome sequencing data and hate-speech detection in Serbian. The stories were then turned into 20 system use cases [8].

The SAIFA design proposes a basic proof-of-concept setup to test the candidate tools together: a web portal, login through home institutions, an access-control module, PostgreSQL with S3 storage, JupyterHub, a model-serving service, a separate bridge to SLURM, and Prometheus/Grafana monitoring. For each part, the setup must show where the data is stored, how the user's identity is passed on, how resource use is charged and logged, how access is checked and what happens when another service fails [8]. The beta version of the gateway is planned for April 2027 and must offer at least one full path for a user: log in, find a resource, run a job and follow its progress.

VII. Conclusion

The main contribution of this paper is a model of an AI gateway with a clear division of responsibilities between the gateway, existing systems and partner Factories. The comparison with Waldur, Open OnDemand and the EFP shows that most new development is needed for the AI-specific functions and the work with linked Factories. Open questions for future work are how gateways will use the EFP for login and allocation, how to track usage across linked Factories, which common metadata to use for models and datasets, and which data rules apply when jobs run in another country.

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